



FeastOS

Restaurant Web Application

Product Requirements Document

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Prepared by: Product Team | Status: Draft → Review

1. Executive Summary

FeastOS is a full-stack, cloud-native restaurant web application designed for modern food-service businesses. It unifies online ordering, real-time table reservations, kitchen operations, loyalty programs, and analytics into a single coherent platform — eliminating the fragmented stack most restaurants cobble together today.

The platform targets three restaurant archetypes: quick-service (QSR), casual dining, and fine-dining establishments. It is built mobile-first, PWA-compliant, and designed to integrate with third-party delivery networks, POS systems, and payment gateways out of the box.

Metric	Year 1 Target
Online order conversion rate	> 4.5 %
Average order processing time	< 90 seconds
Table booking abandonment rate	< 15 %
Customer retention (loyalty)	> 35 % repeat visits
Uptime SLA	99.9 %
Mobile Lighthouse score	> 90 / 100

2. Problem Statement

2.1 Current Pain Points

- Restaurants manage 4–7 disconnected tools (POS, reservation app, delivery aggregator, loyalty platform, analytics).
- No single source of truth for inventory, capacity, or customer data.
- Online menus are static and slow to update — pricing and availability lag behind real conditions.
- Loyalty programs are siloed; customers cannot earn points across dine-in and delivery.
- Kitchen staff receive orders through multiple channels causing duplication errors.
- Restaurant owners lack real-time financial and operational dashboards.

2.2 Opportunity

The global online food delivery market is projected to exceed \$1.6 trillion by 2027. Restaurants that own their digital experience (vs. relying solely on aggregators) retain 3× more margin per order. FeastOS captures that opportunity by giving restaurants a white-label, owned digital storefront with a best-in-class UX.

3. Product Vision & Goals

3.1 Vision Statement

"Give every restaurant — from a 10-seat bistro to a 500-cover chain — the same digital superpowers that Michelin-starred brands and global fast-food chains have, at a fraction of the cost."

3.2 Strategic Goals

- Own-channel revenue: drive $\geq 30\%$ of orders through the restaurant's own app/web within 12 months.
- Operational efficiency: reduce order-to-kitchen latency by 40% vs. manual/phone orders.
- Customer lifetime value: increase repeat visit rate through data-driven loyalty and personalization.
- Multi-location scalability: single dashboard for chains managing up to 500 locations.
- Open ecosystem: REST + Webhook APIs enabling 3rd-party integrations without vendor lock-in.

4. User Personas

■ Diner — 'Amara'

- Age: 28, urban professional, mobile-first
- Wants: fast ordering, real-time tracking, dietary filters (vegan/gluten-free)
- Frustration: re-entering details every visit, no loyalty rewards on web orders
- Goal: seamless omnichannel experience — book, order, pay, earn points in one place

■■■ Kitchen Staff — 'Chidi'

- Age: 35, head chef at a casual dining restaurant
- Wants: single KDS screen showing all orders regardless of channel
- Frustration: paper tickets from phone orders mixed with tablet tickets from aggregators
- Goal: clear priority queue, allergen flags, and prep-time estimates

■ Restaurant Manager — 'Ifeoma'

- Age: 42, operations manager for a 3-location chain
- Wants: real-time revenue dashboard, staff scheduling, inventory alerts
- Frustration: pulls data from 5 different CSV exports every Monday
- Goal: unified analytics with auto-generated weekly reports

■ IT Admin — 'Emeka'

- Age: 38, tech lead overseeing restaurant systems
- Wants: SSO, role-based access, API docs, POS integration
- Frustration: no webhook support, poor API documentation
- Goal: stable REST API with sandbox environment

5. Feature Requirements

5.1 Module Overview — Priority Matrix

Module	Priority	MVP?	Key Capability
Online Ordering (Web + PWA)	P0	Yes	Cart, checkout, real-time status tracking
Digital Menu Management	P0	Yes	Live pricing, categories, photos, dietary tags
Table Reservations	P0	Yes	Real-time slot availability, confirmation email/SMS
Payment Gateway	P0	Yes	Stripe, Paystack, cash-on-delivery, split bills
Kitchen Display System (KDS)	P0	Yes	Live order queue, prep timers, allergen flags
Customer Accounts & Auth	P0	Yes	Email/social login, saved addresses, order history
Loyalty & Rewards Program	P1	No	Points engine, tiers, referral codes, redemption
Real-time Order Tracking	P1	No	Map-based delivery tracking, ETA updates via WS
Push & Email Notifications	P1	No	Transactional + promotional, preference centre
Analytics Dashboard	P1	No	Revenue, top items, peak hours, churn risk
Multi-location Management	P1	No	Centralized menu, per-location overrides, chain KPIs
Staff & Role Management	P2	No	RBAC, shift scheduling, performance metrics
Inventory Tracking	P2	No	Stock levels, low-stock alerts, waste logging
QR Code Dine-in Ordering	P2	No	Scan-to-order at table, split bill by seat
Catering & Event Booking	P3	No	Multi-day orders, deposit payments, custom menus
AI Menu Recommendations	P3	No	Collaborative filtering, upsell engine

5.2 Online Ordering — Detailed Requirements

- Fully responsive PWA with offline-capable service worker (show cached menu when offline).
- Cart persists across sessions via localStorage + server-side sync for logged-in users.

- Real-time item availability: sold-out items auto-hide without page refresh (WebSocket).
- Modifier support: add-ons, remove ingredients, cooking preferences (e.g. 'medium-rare').
- Scheduled orders: place order now, choose delivery/pickup time up to 7 days ahead.
- Group ordering: share a cart link; each member adds their own items, one checkout.
- Dietary filter panel: vegan, halal, gluten-free, nut-free, calorie display.

5.3 Reservation System — Detailed Requirements

- Interactive floor-plan selector: guests choose table zone (indoor/outdoor/bar/private).
- Google Calendar & Apple Calendar ICS export on confirmation.
- Automated SMS/email reminders: 24 hrs and 2 hrs before booking.
- No-show protection: optional card pre-authorisation for bookings of 6+ covers.
- Waitlist management: auto-notify next in queue when a booking cancels.
- Special occasions: birthday/anniversary flags trigger front-of-house notes to staff.

5.4 Loyalty Program — Detailed Requirements

- Points earned: dine-in, online orders, referrals, app reviews, birthday bonus.
- Tier structure: Bronze → Silver → Gold → Platinum with escalating perks.
- Redemption: points for discounts, free items, priority booking, exclusive events.
- Gamification: streaks, challenges ('Order 5 times this month, earn 2× points').
- Referral engine: unique referral codes tracked to source, reward on first order.

5.5 Analytics Dashboard — Detailed Requirements

- Revenue: daily/weekly/monthly P&L, comparison vs. prior period, forecast.
- Menu performance: best/worst sellers, margin per item, attachment rate for add-ons.
- Customer intelligence: new vs. returning, LTV cohorts, churn risk score.
- Operational: average ticket time, order error rate, peak-hour heatmap.
- Export: CSV, PDF, scheduled email delivery to stakeholders.

6. Technical Architecture

6.1 Stack Overview

Layer	Technology / Service	Rationale
Frontend	Next.js 14 (App Router) + TypeScript	SSR/ISR for SEO, PWA support, great DX
Mobile App	React Native (Expo) + TypeScript	Code share with web, OTA updates
UI Components	shadcn/ui + Tailwind CSS	Accessible, headless, fast customization
State Management	Zustand + React Query (TanStack)	Lightweight global state + server state
Backend API	Node.js + Fastify (REST + tRPC)	High throughput, type-safe API contracts
Real-time	Socket.IO over Redis Adapter	Order status, KDS, inventory updates
Database	PostgreSQL (primary) + Redis (cache)	ACID for transactions, sub-ms cache reads
ORM	Prisma ORM	Type-safe queries, automatic migrations
Auth	NextAuth.js + JWT + OAuth2	Social login, magic link, RBAC middleware
File Storage	AWS S3 + CloudFront CDN	Menu images, receipts, QR codes
Payments	Stripe (global) + Paystack (Africa)	Regional payment method coverage
Email/SMS	Resend (email) + Twilio (SMS)	Transactional reliability, webhooks
Search	Meilisearch (self-hosted)	Fast menu search with typo tolerance
Monitoring	Sentry + DataDog + Grafana	Error tracking, APM, custom dashboards
CI/CD	GitHub Actions + Docker + AWS ECS	Blue-green deployments, zero downtime
Infrastructure	AWS (ECS Fargate + RDS + ElastiCache)	Managed, auto-scaling, multi-AZ

6.2 API Design Principles

- RESTful APIs versioned under `/api/v1/`; tRPC for internal type-safe calls.
- OpenAPI 3.1 spec auto-generated and hosted at `/docs`; Postman collection exported.
- Rate limiting: 100 req/min per IP (public), 1000 req/min per authenticated tenant.
- Idempotency keys required on all POST endpoints for payment and order creation.
- Webhook delivery with exponential backoff, HMAC-SHA256 signature verification.

6.3 Security Requirements

- OWASP Top 10 compliance verified via quarterly penetration testing.
- PCI-DSS SAQ-A compliance: no raw card data touches FeastOS servers.
- Data encryption at rest (AES-256) and in transit (TLS 1.3+).
- GDPR / NDPR compliant: data export, right-to-erasure, consent management.
- MFA available for admin accounts; enforced for roles with financial access.

7. UX & Design System

7.1 Design Principles

- Mobile-first: all flows designed for 375px viewport, enhanced for larger screens.
- Speed: Largest Contentful Paint < 2.5s, Cumulative Layout Shift < 0.1.
- Accessibility: WCAG 2.1 AA — keyboard navigation, screen reader support, 4.5:1 contrast.
- Consistency: shared Figma design token library drives both web and mobile components.
- Localisation: i18n-ready from day one; RTL layout support for Arabic/Hebrew markets.

7.2 Key User Flows

Flow A — First-time Online Order (Guest)

- Land on restaurant homepage → Browse menu (filter by category/dietary) → Add to cart → Enter details (name, address, phone) → Choose payment → Order confirmed → Track in real-time.
- Total steps to first order: ≤ 6 taps on mobile.

Flow B — Returning Customer (Logged In)

- Open app → Tap 'Reorder' on last order → Confirm address → Pay with saved card → Done.
- Time to reorder: < 30 seconds.

Flow C — Table Reservation

- Select date/party size → View available times → Choose time slot → (Optional) pick zone → Enter name/phone → Confirm → Receive SMS + calendar invite.

8. Non-Functional Requirements

Category	Requirement	Metric
Performance	API p95 response time	< 200ms
Performance	Page load (mobile, 4G)	< 3 seconds
Scalability	Concurrent users per restaurant	1,000+
Scalability	Platform-wide concurrent users	100,000+
Availability	Uptime SLA (platform)	99.9% (< 8.7 hrs/yr)
Availability	Uptime SLA (payment flow)	99.95%
Recovery	RTO (major outage)	< 30 minutes
Recovery	RPO (data loss max)	< 5 minutes
Security	SSL/TLS everywhere	TLS 1.3 minimum
Security	Auth token expiry	15 min access / 7 day refresh
Compliance	GDPR data export	< 72-hour response
Accessibility	WCAG level	2.1 AA

9. Third-Party Integrations

Integration	Provider(s)	Purpose
Payment Processing	Stripe, Paystack, Flutterwave	Card, bank transfer, USSD, mobile money
Delivery Logistics	Google Maps Platform	Distance calc, route display, geofencing
SMS Notifications	Twilio, Termii (Africa)	OTP, order status, reminders
Email	Resend, SendGrid (fallback)	Transactional emails, marketing campaigns
POS Systems	Square, Toast, Revel (REST API)	Sync orders, menu, sales data
Social Login	Google, Facebook, Apple	OAuth2 authentication
Analytics	Google Analytics 4, Mixpanel	Funnel analysis, event tracking
Review Platforms	Google Reviews API	Display ratings, prompt review post-order
Accounting	QuickBooks, Xero (webhooks)	Auto-sync daily revenue and expenses
Delivery Aggregators	Glovo, Bolt Food (webhooks)	Receive orders from aggregators into KDS

10. Monetization & Pricing Model

Plan	Price	Locations	Key Features
Starter	\$49/mo	1	Online menu, ordering, reservations, basic analytics, email support
Growth	\$129/mo	1–3	All Starter + loyalty, QR dine-in, push notifications, KDS, priority support
Pro	\$299/mo	1–10	All Growth + multi-location dashboard, API access, inventory, SLA 99.9%
Enterprise	Custom	Unlimited	All Pro + dedicated infra, custom integrations, SLA 99.95%, CSM

- Transaction fees: 0% on Starter; 0.5% on Growth; 0% on Pro+ (restaurant uses own payment keys).
- Add-ons: AI recommendations (\$29/mo), advanced marketing automation (\$49/mo), white-label app (\$199 setup + \$49/mo).

11. Success Metrics & KPIs

OKR	Key Result	Target (12 months)
Grow platform adoption	Paying restaurant accounts	500 restaurants
Increase diner engagement	Monthly active diners (MAD)	50,000 MAD
Drive owned-channel revenue	% orders via FeastOS vs. aggregator	> 30%
Improve operational quality	Order accuracy rate	> 98%
Build loyalty	Loyalty program enrolment	> 40% of registered users
Maintain reliability	Platform uptime	> 99.9%
Ensure satisfaction	NPS score — restaurants	> 55
Ensure satisfaction	NPS score — diners	> 60

12. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Payment gateway downtime	Low	Critical	Multi-provider fallback (Stripe → Paystack); circuit breaker pattern
POS API breaking changes	Medium	High	Versioned adapter layer; contract tests in CI pipeline
GDPR / NDPR non-compliance	Low	Critical	Legal review at each feature launch; DPA in place
Mobile performance on low-end devices	Medium	High	Progressive enhancement; < 200KB initial JS bundle
Aggregator disintermediation	High	Medium	Loyalty incentives for direct orders; own-channel pricing discounts
Fraudulent orders / chargebacks	Medium	High	Stripe Radar + custom velocity rules; phone verification at checkout
Team velocity / scope creep	High	Medium	Fixed-scope sprints; MVP gating; weekly stakeholder demos

13. Open Questions & Decisions

- [Decision needed] Should MVP include real-time driver tracking or is ETA text sufficient?
- [Decision needed] Self-hosted Meilisearch vs. Algolia — cost vs. operational overhead trade-off.
- [Research needed] Which African POS systems (e.g. Aloha, Sage) have the highest penetration in target markets?
- [Legal review] Token-based loyalty points — do they constitute a financial instrument in NG/ZA jurisdictions?
- [UX research] Test 3-step vs. 5-step checkout flow with real users before committing to MVP checkout.
- [Architecture] Monolith-first or microservices from day one? Recommendation: modular monolith, extract services post-PMF.

14. Appendix

14.1 Glossary

- KDS — Kitchen Display System: screen in the kitchen showing live order queue.
- PWA — Progressive Web App: web app installable on mobile with offline support.
- RBAC — Role-Based Access Control: permissions tied to user roles, not individuals.
- RTO — Recovery Time Objective: max acceptable downtime after an incident.
- RPO — Recovery Point Objective: max acceptable data loss in time units.
- LTV — Customer Lifetime Value: predicted revenue from a customer over their relationship.

14.2 Document History

Version	Date	Author	Changes
0.1	Jan 2026	Product Team	Initial draft — scope & personas
0.5	Feb 2026	Product + Eng	Tech stack finalised; NFRs added
0.9	Mar 2026	All stakeholders	Risk register; monetisation model
1.0	Apr 2026	Product Lead	Final review; approved for engineering

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